DS-112 Capstone Project Report

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Preprocessing

- At the beginning of my code, after all the imports, I seeded the random number generator with my N-number, N17684158, using np.random.seed(17684158).
- I excluded professors with fewer than 5 ratings to reduce noise from extreme sample averages. Choosing 5 as the threshold still retains a large dataset for analysis, which ensures that we are not losing too much data.
- Filtering out professors with fewer than 5 ratings already removed a large number of rows that contain missing values (NaN in number of ratings column means that information provided by this row would be very limited), so I would apply data cleaning specific to questions (drop rows that contain missing values in columns relevant to the question), which would be more flexible.

Question 1

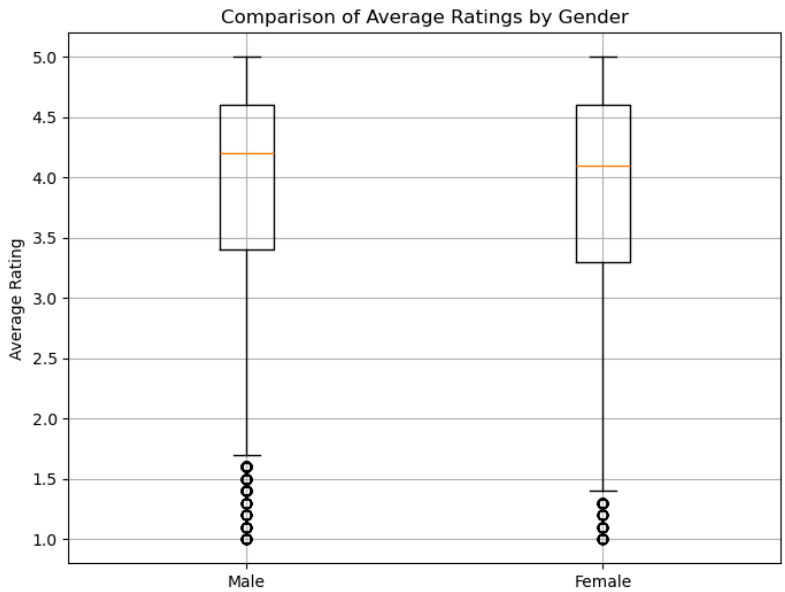
For Q1, I examined whether male professors receive significantly higher ratings than female professors. Since my preprocessing step already filtered out professors with fewer than 5 ratings, along with those who do not have any ratings at all, the remaining dataset contained no missing values in the average rating column. I also checked the gender columns (male and female), and the sums of the missing values in these two columns are both zero, which means that there are no missing values in these two columns. Therefore, there is no additional data cleaning for this analysis.

Since we could not assume normality and homogeneity of variance, I used Mann–Whitney U test to compare the average ratings of male and female professors:

- H_0: There is no difference in average ratings between male and female professors.
- H_1: There is a difference in average ratings between male and female professors.
- Median rating of male professors: 4.2
- Median rating of female professors: 4.1
- U statistic = 50901762.5
- p-value = 0.00084

Since $p < 0.005$ (our alpha level), our null hypothesis is rejected, and I conclude that the difference in average ratings between male and female professors is statistically significant and unlikely to be due to chance. There is evidence of a pro-male gender bias in this dataset.

Figure: A boxplot comparing the average ratings of male and female professors



Question 2

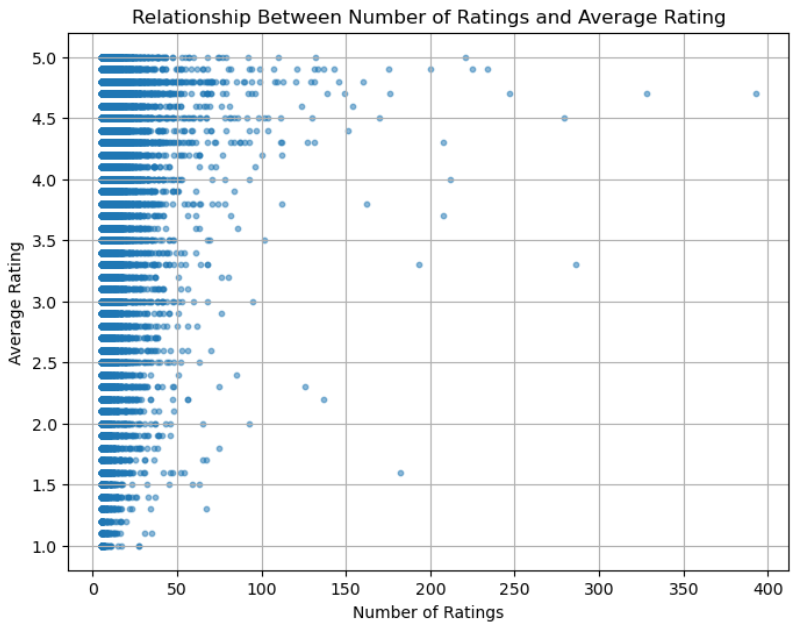
For Q2, I tested whether teaching experience is associated with teaching quality by finding the correlation between the number of ratings and the average ratings. The two relevant columns were already cleaned at the beginning, so no additional data cleaning needed.

Since the scatterplot seems show a nonlinear relationship, I ran a Spearman's rank correlation test to find if there is a linear relationship between the two variables:

- H_0: There is no relationship between number of ratings and average rating.
- H_1: There is a relationship between number of ratings and average rating.
- rho = 0.028
- p-value = 6.92×10^{-6}

Result is statistically significant at the $\alpha = 0.005$ level, which rejects the null hypothesis, suggesting that there is a very weak positive association between teaching experience and teaching quality. Although results show evidence that there is a relationship between the two variables, the correlation is too weak that the effect is not practically meaningful.

Figure: A scatterplot showing the relationship between number of ratings and average rating

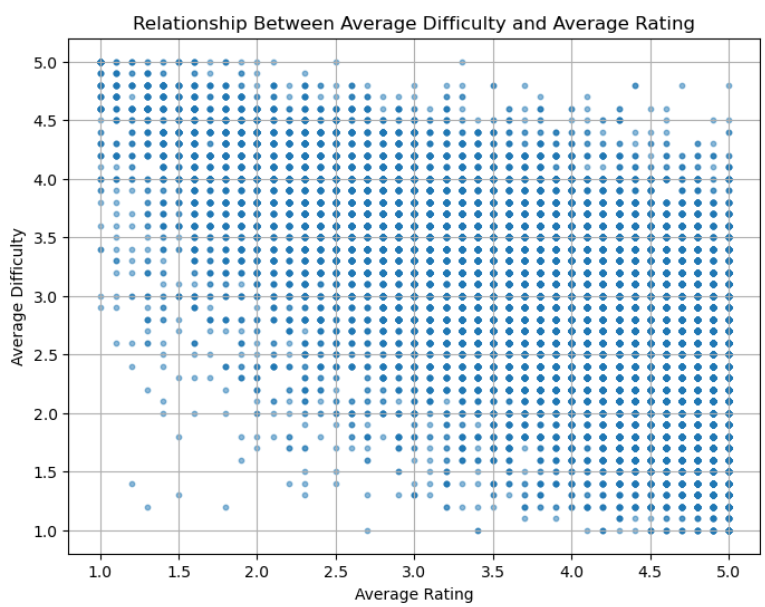


Question 3

For Q3, I examined the relationship between average rating and average difficulty. The two relevant columns were already cleaned at the beginning, so no additional data cleaning needed.

I ran a Pearson correlation test to find the correlation between the two variables. The correlation coefficient is r = -0.619, suggesting a moderately strong negative correlation: higher ratings are associated with lower difficulty.

Figure: A scatterplot showing the relationship between average rating and average difficulty



Question 4

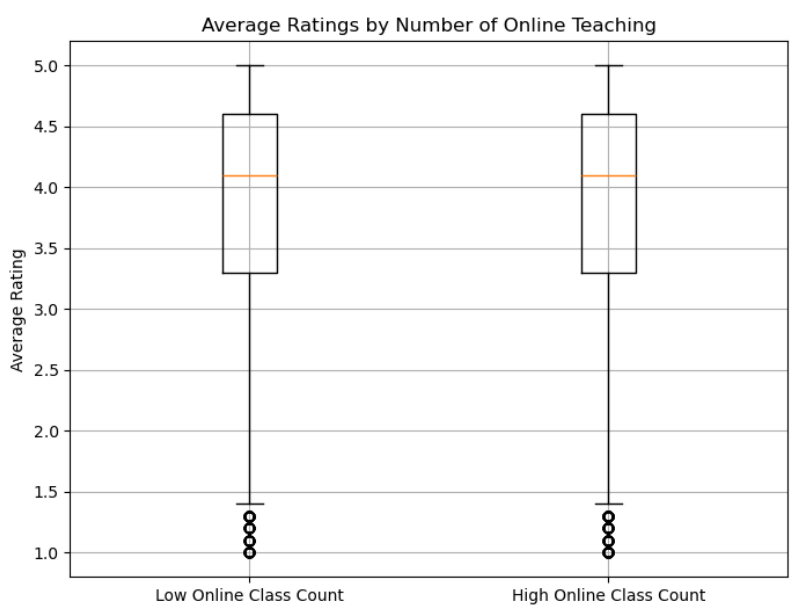
For Q4, I examined whether professors who teach a lot of classes in the online modality receive higher or lower ratings than those who don't. I checked that the online class ratings count column does not contain any missing values, so no additional data cleaning needed.

Based on the number of online class ratings, I set the top 20% of professors to represent those who teach a lot of online classes and the bottom 20% to represent those who teach very few. Since the distributions of the two groups of ratings we are evaluating are potentially skewed and different, I conducted a Mann–Whitney U test for this question as it compares median rather than mean:

- H_0: There is no difference in the distribution of average ratings between professors who teach many online classes and those who teach very few.
- H_1: There is a difference in the distribution of average ratings between professors who teach many online classes and those who teach very few.
- Median rating of low online class count group: 4.1
- Median rating of high online class count group: 4.1
- U statistic = 259558200.5
- p-value = 0.331

Result shows that the p-value > 0.005, suggesting no statistically significant difference. I conclude that teaching many online classes does not significantly affect how students rate professors.

Figure: A boxplot comparing average ratings of professors who teach very few versus many online classes

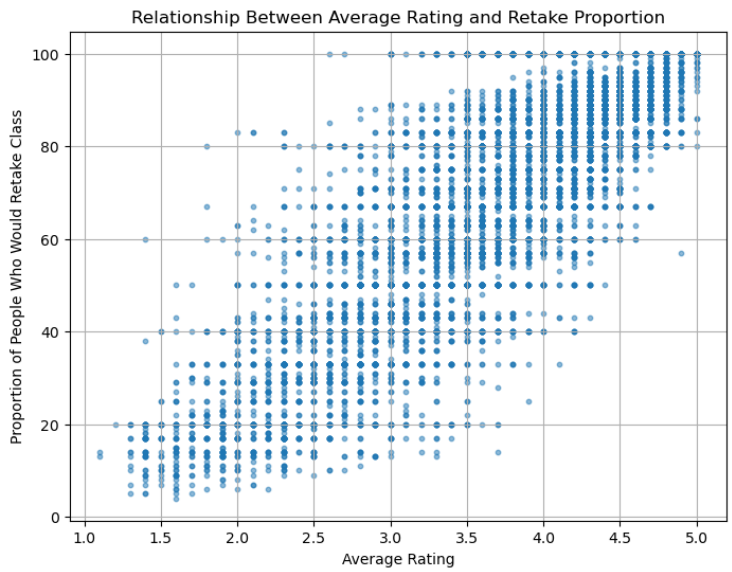


Question 5

For Q5, I tested the relationship between average rating and the proportion of people who would retake the class. The retake proportion column has many missing values that were not cleaned at the beginning, so I dropped these rows to clean the data.

I ran a Pearson correlation test to find the correlation between the two variables. The correlation coefficient is r = 0.88, suggesting a very strong positive correlation: higher ratings are associated with larger proportion of people who would take the class again.

Figure: A scatterplot showing the relationship between average rating and the proportion of people who would retake the class



Question 6

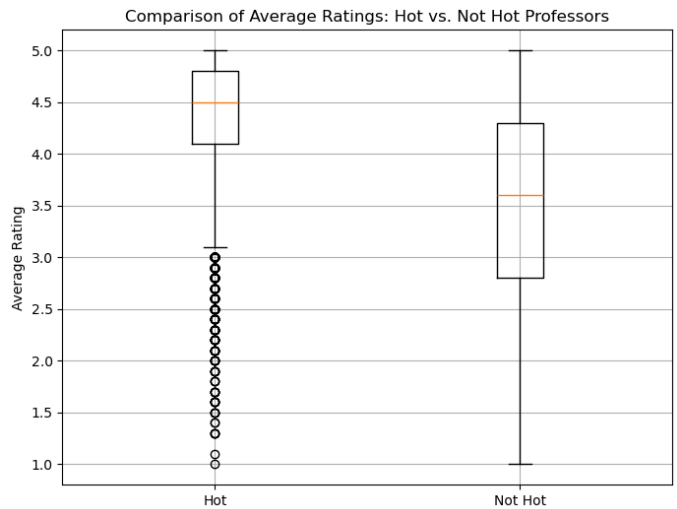
For Q6, I examined whether professors judged as “hot” receive higher ratings compared to those who are not. I checked that the “pepper” column does not contain any missing values after the initial data cleaning, so no additional data cleaning needed.

Since the distributions of the two groups of ratings we are evaluating are potentially skewed and different, I conducted a Mann–Whitney U test for this question as it compares median rather than mean:

- H_0: There is no difference in the distribution of ratings for hot and not hot professors.
- H_1: There is a difference in the distribution of ratings hot and not hot professors.
- Median rating of "hot" professors: 4.5
- Median rating "not hot" professors: 3.6
- U statistic = 121910581.5
- p-value = 0.0

Result is statistically significant at $\alpha = 0.005$ level, so we reject the null hypothesis, suggesting that hot professors receive higher ratings than those who are not.

Figure: A boxplot comparing the average ratings of "hot" and "not hot" professors

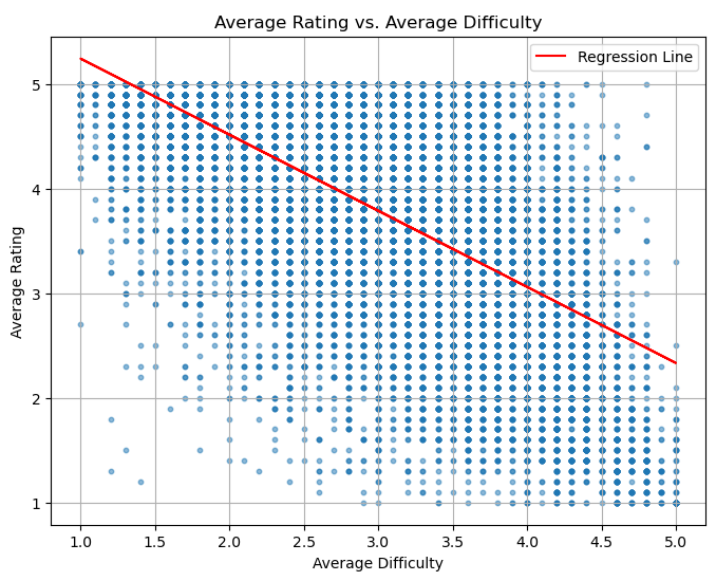


Question 7

For Q7, I built a linear regression model to predict a professor's average rating based on their average difficulty only. Average rating and average difficulty columns were cleaned already, so no additional data cleaning needed.

- The model achieved an R^2 of 0.383, indicating that difficulty explains about 38.3% of the variance in ratings.
- The RMSE is 0.744, meaning the model's typical prediction error is approximately 0.744.

Figure: A scatterplot showing average ratings vs. average difficulty with linear regression line



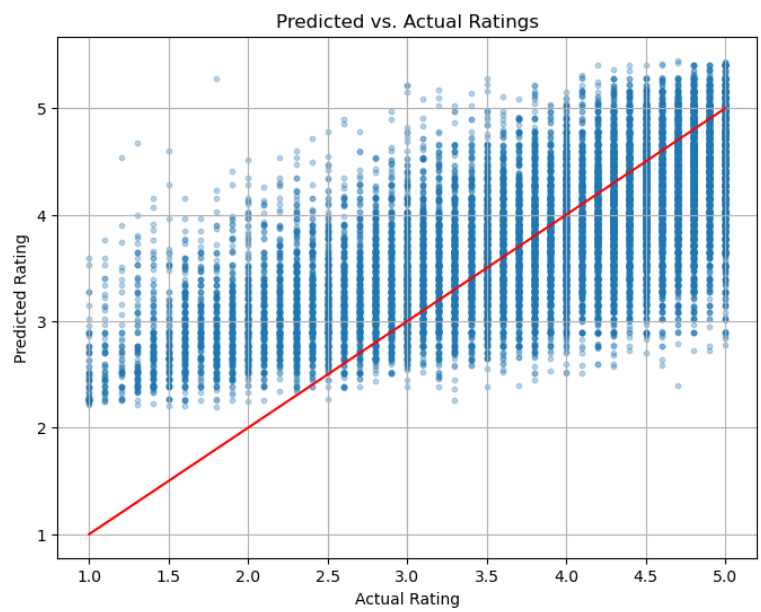
Question 8

For Q8, in order to avoid collinearity, I excluded the gender columns, male and female, since their relationship is a perfect negative correlation. By making a correlation matrix, I found that retake proportion has a moderately negative correlation, r = –0.52, with average difficulty, suggesting some collinearity. Also, the retake proportion column contains many missing values, so I decided to exclude it as well.

- The model achieved an R^2 of 0.485, indicating that difficulty explains about 48.5% of the variance in ratings.
- The RMSE is 0.679, meaning the model's typical prediction error is approximately 0.679.

This expanded model offers better predictive performance than the difficulty only model, with higher explanatory power, R^2, and lower prediction error, RMSE.

Figure: A scatterplot showing predicted vs. actual average ratings using a linear regression model with four predictors



Question 9

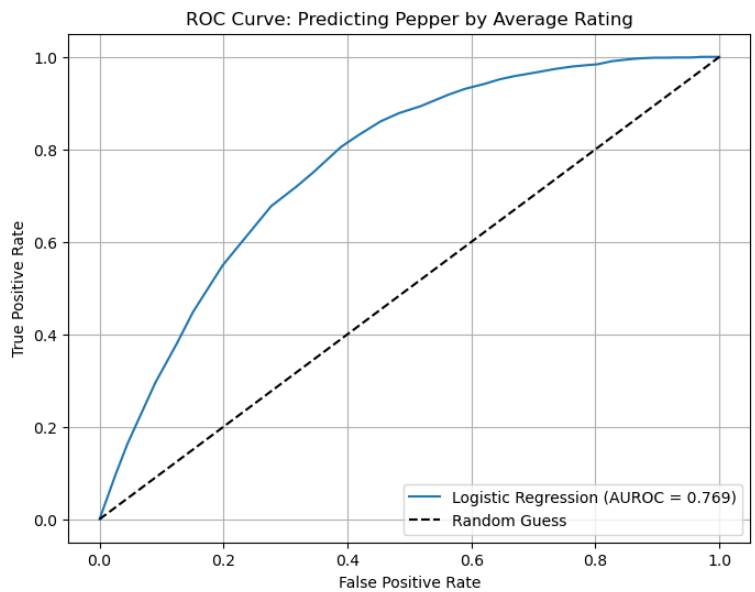
For Q9, I built a logistic regression model to predict whether a professor receives a “pepper” using only their average rating. Class imbalance was addressed using class weighting.

Metrics:

- Accuracy: 0.691
- Precision: 0.595
- Recall: 0.805
- AUROC: 0.769

The model performs moderately well with accuracy score of 0.691. Recall is high, meaning the model is good at identifying actual “hot” professors. Precision is lower, suggesting some false positives. AUROC = 0.769 shows that the model is good at ranking hot vs. not hot professors based on average rating.

Figure: ROC curve for a logistic regression model predicting whether a professor receives a “pepper” using average rating



Question 10

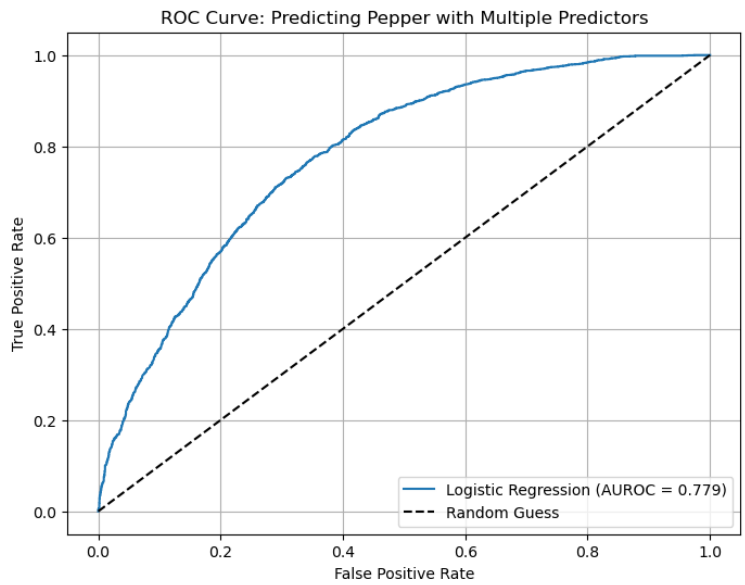
For Q10, I also excluded the gender columns and the retake proportion column for the same reasons as in Q8. Class imbalance was addressed using class weighting.

Metrics:

- Accuracy: 0.697
- Precision: 0.605
- Recall: 0.785
- AUROC: 0.779

Compared to the average ratings only model, this expanded model does not show significant improvements through these metrics. This suggests that adding other predictors beyond average rating does not significantly enhance the model’s performance to predict "pepper".

Figure: ROC curve for a logistic regression model predicting whether a professor receives a "pepper" using four predictors (average rating, average difficulty, number of ratings, and online class count)



Extra Credit – Major and Average Rating

A common stereotype: STEM courses tend to be more difficult and rigorous, and they are more likely to receive lower ratings. In this analysis, I explored whether professors in STEM fields tend to receive lower student ratings compared to those in non-STEM fields.

Average difficulty might also be a predictor here, but I decided not to use it because we already knew from Q3 that average difficulty and average rating are negatively correlated. Therefore, as we are focusing on STEM vs. non-STEM, including average difficulty as one of the predictors would weaken our emphasis on rating difference by major.

In the preprocessing stage, I filtered out professors with less than 5 ratings. In order to align the qualitative data with the numeric data, I filtered out the same rows in the qualitative dataset. Average ratings column was already cleaned at the beginning. I checked that the majors column contains 5188 missing values, so I dropped these rows.

For this analysis, I first classified each professor's academic field as STEM or non-STEM, using the qualitative dataset. I then constructed a simple linear regression model to predict average rating from a binary indicator of whether a professor's academic field was in STEM.

- Coefficient = -0.011
- $R^2 = 2.42$
- RMSE = 0.948

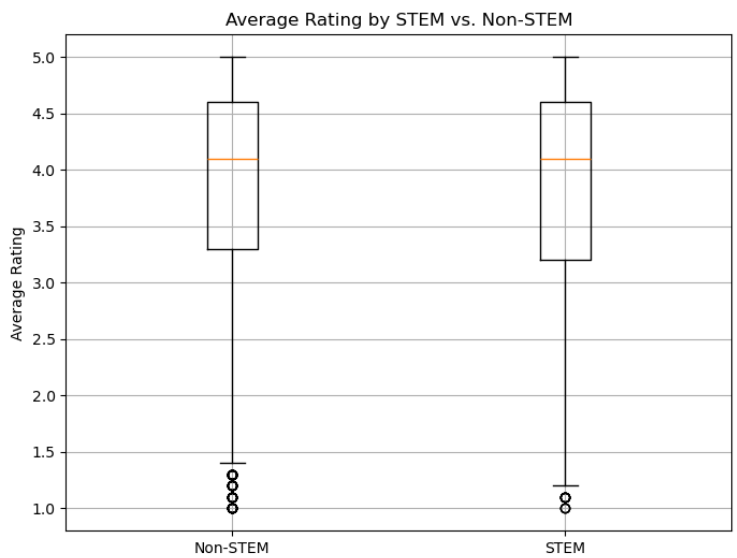
The model shows that the coefficient is slightly negative, suggesting that professors in STEM fields are rated just slightly lower on average than professors in non-STEM fields. The R^2 value is extremely low, indicating that whether a professor is in a STEM field explains virtually none of the variation in ratings. The RMSE is very high, demonstrating the model's poor predictive performance.

The results do not align with the common stereotype that we are testing. To further examine this, I conducted a Mann–Whitney U test to see if there is significant difference in average rating distributions between STEM and non-STEM professors.

- U statistic = 10615639.5
- p-value = 0.90

$p < 0.005$ shows that the results are not statistically significant, meaning that the observed difference in average rating distributions between STEM and non-STEM professors is likely occurred by chance.

Figure: A boxplot comparing average professor ratings between non-STEM and STEM fields



Overall, the results from both the linear regression model and the Mann–Whitney U test provide strong evidence against the common stereotype that professors in STEM fields are usually rated lower by students. These findings suggest that, within this dataset, academic field alone does not meaningfully associate with student evaluations, and the stereotypical perception may be overstated or influenced by other unmeasured factors.